

# System Reliability and Free Riding

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# Introduction

- Reliability depends on effort of many people, so it is a public good
- Individuals may free ride on others
- Temptation to free ride depends on benefits, costs, other agents' effort, technology and liability.
- Baseline cases

**Total effort.** Reliability depends on the sum of the efforts exerted by the individuals.

**Weakest link.** Reliability depends on the minimum effort.

**Best shot.** Reliability depends on the maximum effort.

# Notation

- $x_i$ : effort of agent  $i$
- $c_i x_i$ : cost of effort to agent  $i$
- $P(x_1, x_2)$ : probability of successful operation (increasing, concave)
- $v_i$ : value to  $i$  of successful operation
- So agent 1 cares about  $v_1 P(x_1, x_2) - c_1 x_1$

# Model

- Individual optimization:

$$\max_{x_1} v_1 P(F(x_1, x_2)) - c_1 x_1$$

$$\max_{x_2} v_2 P(F(x_1, x_2)) - c_2 x_2$$

- Social optimum:

$$\max_{x_1, x_2} (v_1 + v_2) P(F(x_1, x_2)) - c_1 x_1 - c_2 x_2$$

Technology:

- Total effort:  $F(x_1, x_2) = x_1 + x_2$
- Weakest link:  $F(x_1, x_2) = \min\{x_1, x_2\}$

# Total effort

$$\max_{x_1} v_1 P(x_1 + x_2) - c_1 x_1$$

FOC:  $v_1 P'(x_1 + x_2) = c_1$  for  $x_1 > 0$ .

Defining  $G(\cdot)$  as inverse of  $P'(\cdot)$ , we have

$$x_1 + x_2 = G(c_1/v_1)$$

Or

$$x_1 = \max\{G(c_1/v_1) - x_2, 0\}.$$

Similarly

$$x_2 = \max\{G(c_2/v_2) - x_1, 0\}.$$

# Equilibrium

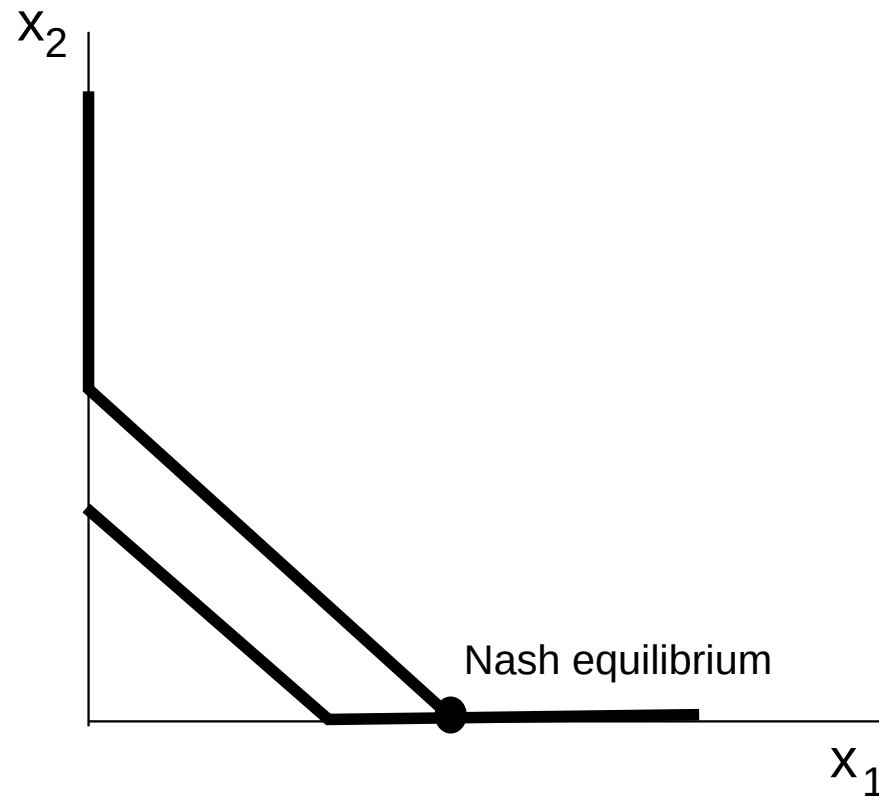


Figure 1: Nash equilibrium in total effort case.

# Social optimum

FOC are

$$(v_1 + v_2)P'(x_1 + x_2) \leq c_1$$

$$(v_1 + v_2)P'(x_1 + x_2) \leq c_2$$

with one of them as an equality, so,

$$x_1 + x_2 = G(\min\{c_1/(v_1 + v_2), c_2/(v_1 + v_2)\})$$

# Weakest link

Agent 1's problem:

$$\max_{x_1} v_1 P(\min(x_1, x_2)) - c_1 x_1.$$

Agent 1 wants to match agent 2's contribution, up to the point where

$$v_1 P'(\bar{x}_1) = c_1$$

# Weakest link equilibrium

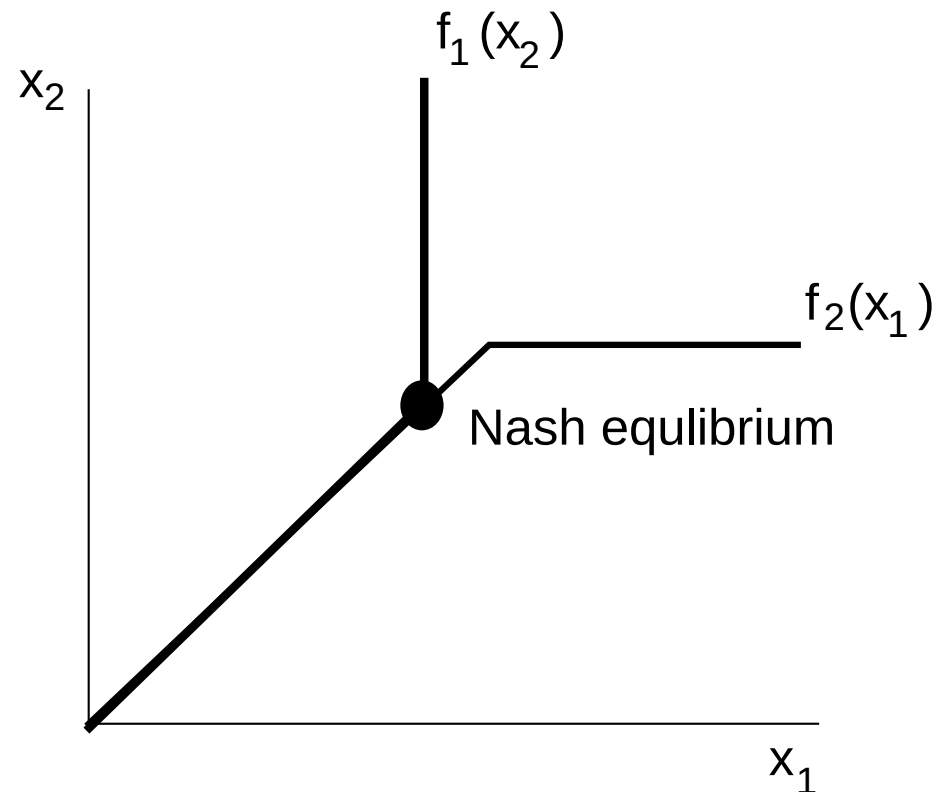


Figure 2: Nash equilibrium in weakest link case.

# Social optimum for weakest link

$$\max_{x_1, x_2} (v_1 + v_2)P(x) - (c_1 + c_2)x$$

yields

$$x^* = G((c_1 + c_2)/(v_1 + v_2))$$

Note that weakest link case is much more costly than total effort!

# Summary of results

**Fact 1** *In the case of total effort, system reliability is determined by the agent with the highest benefit-cost ratio. All other agents free ride on this agent.*

**Fact 2** *In the weakest-link case, system reliability is determined by the agent with the lowest benefit-cost ratio.*

**Fact 3** *In the total effort case, there is always too little effort exerted in the Nash equilibrium as compared with the optimum. Furthermore, when  $v_2/c_2 > v_1/c_1$  but  $c_1 < c_2$ , the “wrong” agent exerts the effort.*

# Results, continued

**Fact 4** *The probability of success full operations in the socially optimal solution is always lower in the case of weakest link that in the case of total effort.*

**Fact 5** *With non-identical agents, systems will become increasingly reliable as the number of agents increases in the total efforts case, but increasingly unreliable as the number of agents increases in the weakest link case.*

**Fact 6** *In the total efforts case with identical agents, the Nash outcome remains constant as the number of agents is increased, but the socially optimal amount of effort increases.*

# Results, continued

**Fact 7** *In the weakest-link case with identical agents, the socially optimal reliability and the Nash reliability are identical, regardless of the number of agents.*

**Fact 8** *A fine (or strict liability) equal to the costs imposed on the other agents should be imposed on the agent who has the lowest cost of reducing the probability of failure.*

**Fact 9** *In the case of weakest link, strict liability is not adequate in general to achieve the socially optimal level of effort, and one must use a negligence rule to induce the optimal effort.*

# Adversaries

Attacker and defender:

$$(1) \quad \max_x \quad vP(x - y) - cx$$

$$(2) \quad \max_y \quad w[1 - P(x - y)] - dy.$$

Reaction functions:

$$(3) \quad x = \min\{\max\{G(c/v) + y, 0\}, \hat{x}\}$$

$$(4) \quad y = \min\{\max\{G(d/w) - x, 0\}, \hat{y}\}.$$

where  $\hat{x}, \hat{y}$  are maximal possible efforts.

# Adversary equilibrium

**Cost/benefit of defender  $<$  attacker** Attacker gives up and defender does just enough to keep him at bay.

**Cost/benefit of defender  $>$  attacker** Defender has to go all out, and the attacker pushes to keep him there.

# Adversaries

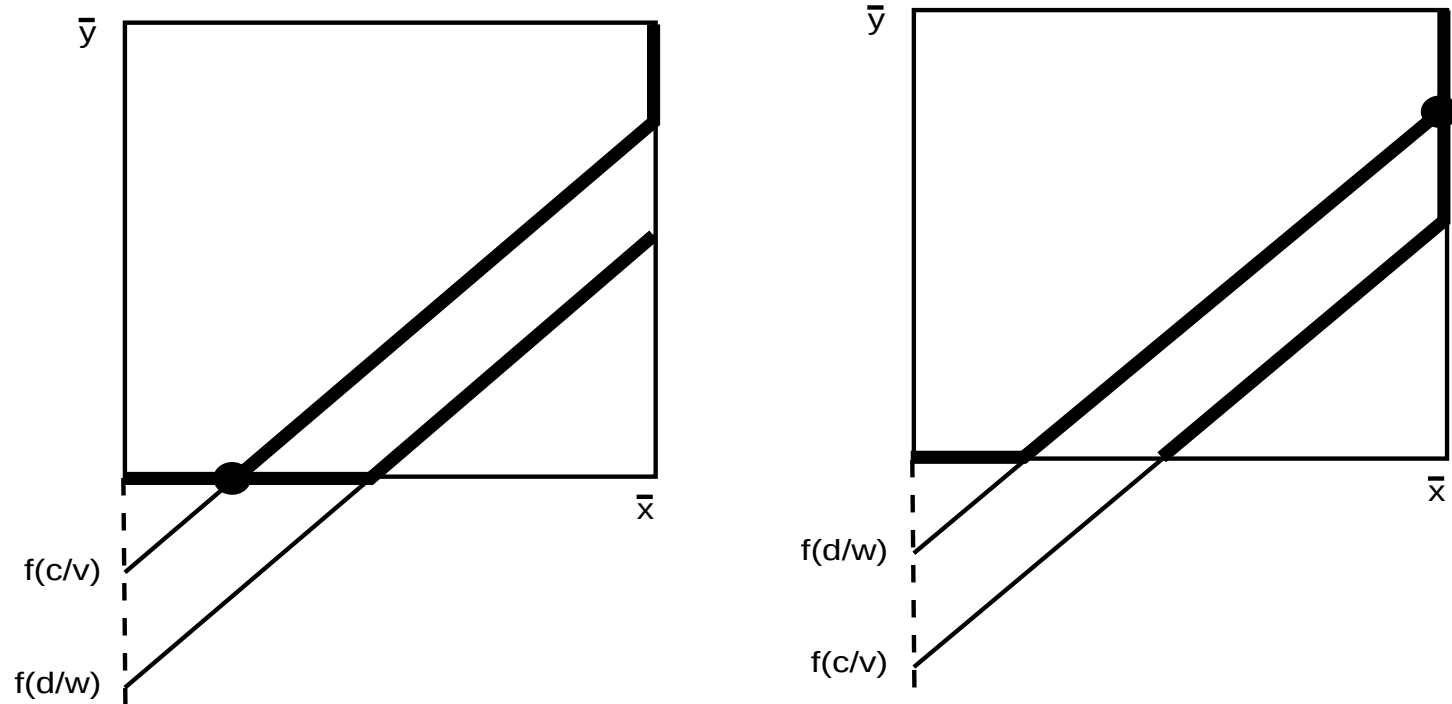


Figure 3: Reaction functions in adversarial case.